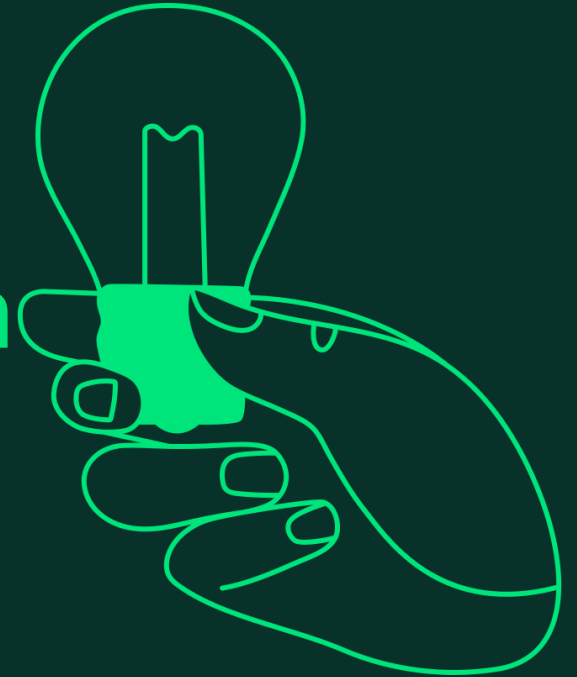


Rubber, meet the Road: **At the interface of Statistical Design and Clinical Operation**



Dooti Roy, Head of Innovation, Clinical Data Science and Methodology
On Behalf of
Efficiency+ Sub-team of Design and Operation

Disclaimer

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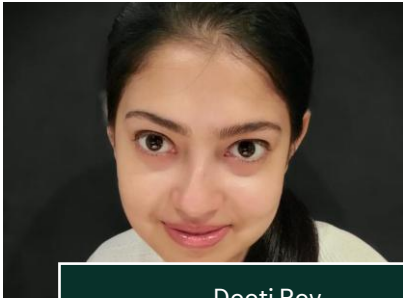
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Dooti Roy – Short Bio

Dr. Dooti Roy is Executive Director of Global Biostatistics and Data Sciences at Boehringer Ingelheim, where she serves as Head of Innovation for Clinical Data Science and Methodology (CDSM) and U.S. Head of CDSM. She leads innovation across statistics, statistical programming, and data science, driving the adoption of scientific and technological advances to accelerate high-quality drug development. A passionate leader, mentor, and statistician, she is committed to building future scientific talent and advancing data-driven, equitable solutions in healthcare. Dr. Roy also holds an adjunct faculty appointment at the University of Connecticut and actively contributes to various cross-industry initiatives and professional associations.

Sub-Team Members



Dooti Roy
(Statistician, Boehringer Ingelheim)



Angela Zhu
(Statistician, Boehringer Ingelheim)



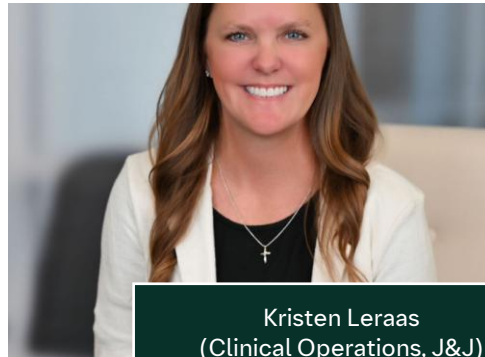
Kristen Daniels
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Kristen Leraas
(Clinical Operations, J&J)



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Kristel Rens
(Drug Supply, J&J)

A brilliant design no site can run delivers zero value

Objective

Inspire differently informed clinical trial designs where optimization extends beyond power and sample size

Approach

Forging a deeper, mutual understanding of role and impact of statistical trial design on partnering functions

Partners in Focus

Clinical Operations, Drug Supply and IRT (Interactive Response Technology)



Statistical trial design as part of a BIGGER picture

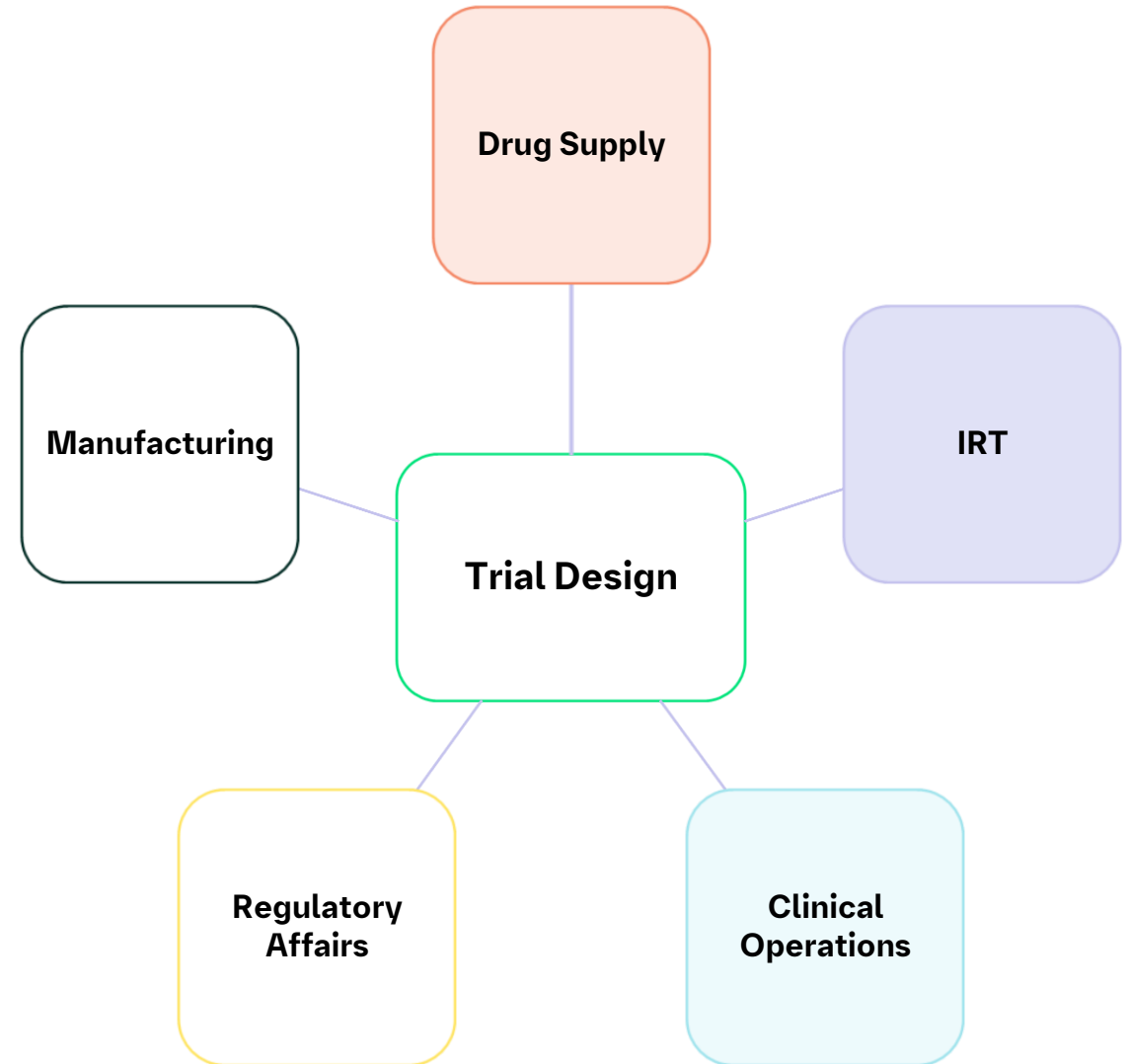
Status quo

In a recent systematic literature review¹ of all published studies between 2010 and 2020, out of 1,263 screened trials, only 317 were adaptive in nature. Of those, it was found that only 8.5% of trials (27/317) reported using a seamless 2-3 design.

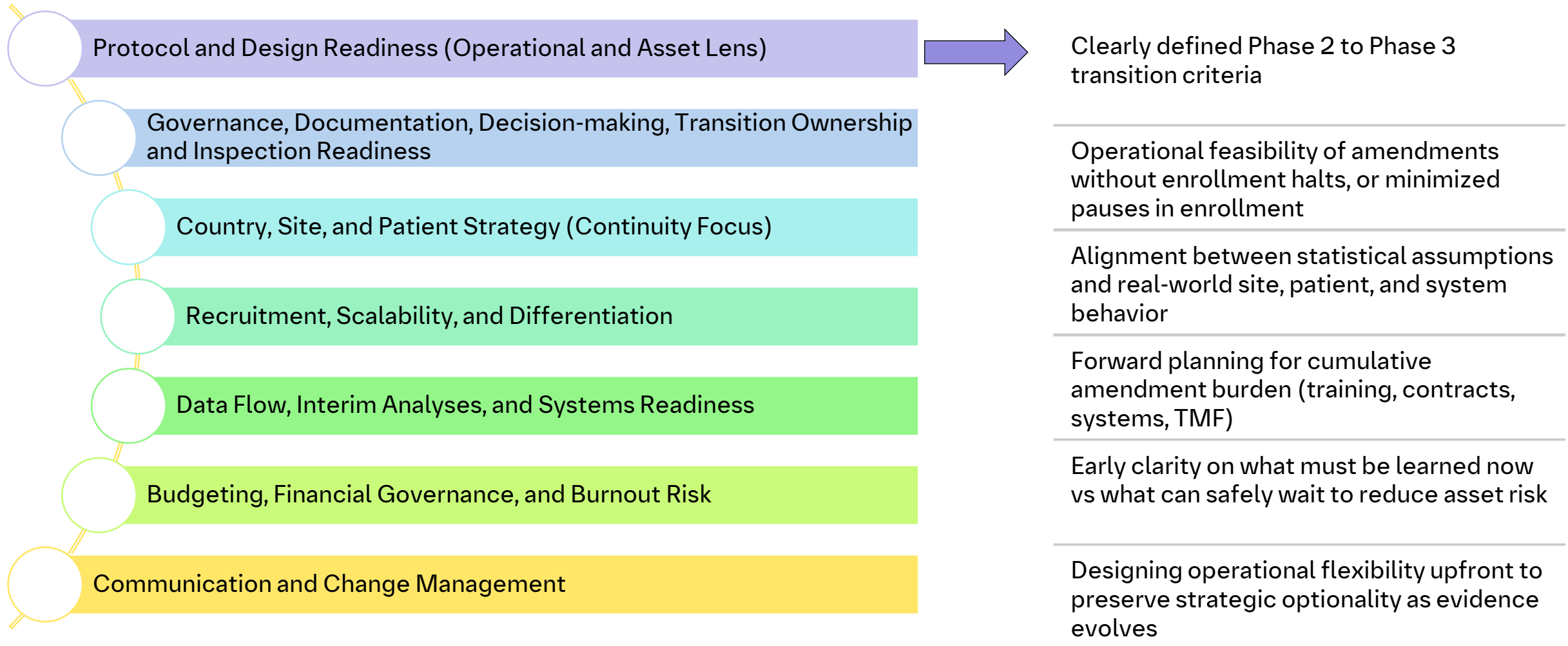
Statisticians often design innovative (adaptive) trials without fully understanding the impact of such designs on other functions such as clinical operations, manufacturing, regulatory affairs, drug supply or IRT (Interactive Response Technology).

Deepening mutual understanding can help improve internal acceptance, adoption and ease implementation of such designs.

1: [Adaptive designs in clinical trials: a systematic review-part I](#) | BMC Medical Research Methodology | Springer Nature Link



Considerations of a seamless 2-3 design on Clinical Operations (*teaser*)



Workstream Outputs

- **Manuscript:** End of 2026: Manuscript under preparation summarizing connectivity, and impact of implementation of seamless 2-3 design on operations, drug supply and IRT
- **Organizational Cross-functional upskilling resource:** The manuscript (and the associated research) will be used to generate content summaries which will improve biostatisticians' cross-functional understanding within their organizations.
- **Public Education and Awareness:** Ongoing including conference presentations and webinars

Thank *you* for listening!